

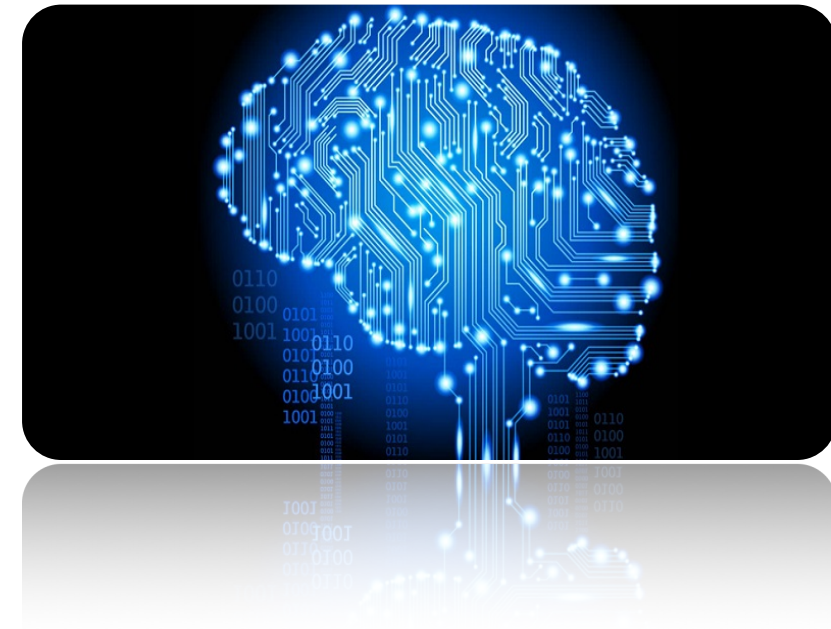
Cognitive Computing

Introduction to Natural language processing (NLP)

Role of NLP in Cognitive System

- **Cognitive System**

- *Mimic human-like reasoning and decision making*
- *Generate and Score Hypothesis*
- *Alternative answers – each with a degree of confidence*

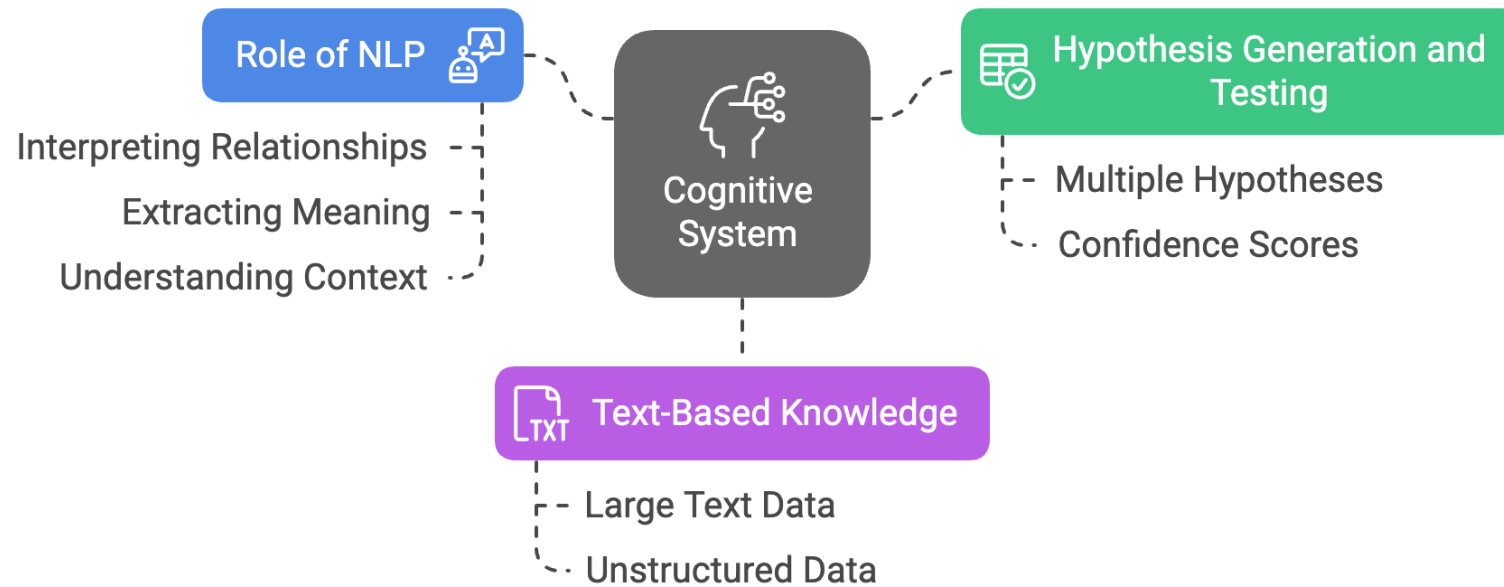


Cognitive system vs other data-driven techniques

- **Capability to manage, understand, and analyze unstructured data in context with the questions being asked. (rely on large amount of text data)**
- **Unstructured information must be parsed and tagged to find the elements of meaning**
- **Understand meaning and intent behind text**
- **Extensive NLP for human like responses**
- **E.g.: IBM Watson, ChatGPT**
- **Works with Structured Data (like tables) and relies on schemas to add context and meaning to data**
- **Identifies patterns but lacks deeper understanding**
- **Limited NLP**
- **SQL-based systems, basic ML models**

Role of NLP

Cognitive System Components: Hypothesis, Knowledge, and NLP



Role of NLP in Cognitive System (contd..)

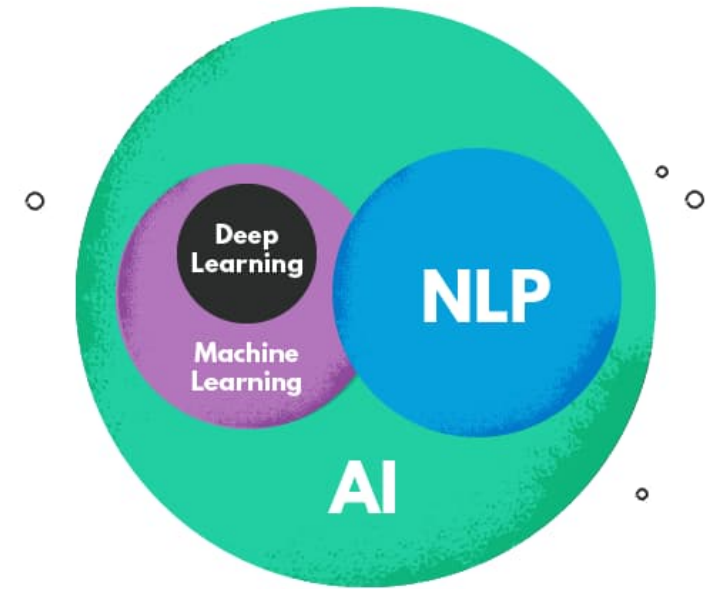
- Tools for this process of identifying the meaning of the individual words include categorization, ontologies, tagging, catalogs, dictionaries, and language models (NLP).
- This is where **Natural Language Processing (NLP) techniques** come in.

Remember - Not all unstructured data is text.

There is a requirement in some cognitive computing systems to support images, video, speech, and sensor data, depending on how the data will be used.

Natural Language Processing (NLP)

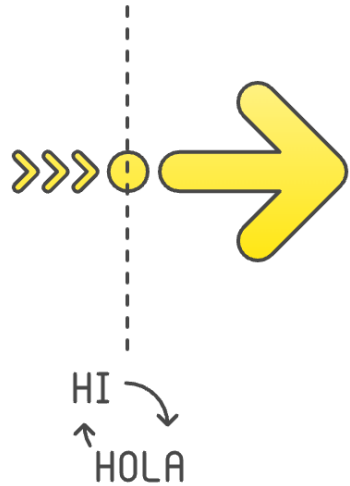
- A field of artificial intelligence (AI) that enables computers to understand, interpret, and generate human language.
- It combines linguistics, machine learning, and computational techniques to process and analyze large amounts of text and speech (unstructured) data.
- NLP – Set of techniques that extract meaning from text



THE HISTORY OF NLP

Early Attempts at Language Transformation

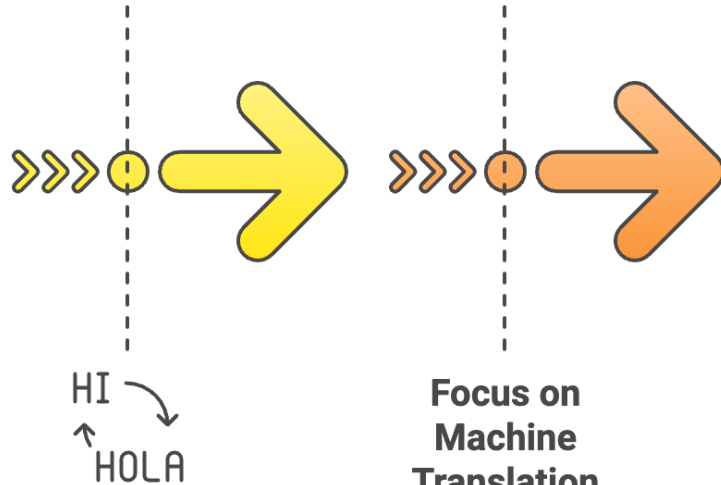
Historians note 17th-century efforts



THE HISTORY OF NLP

Early Attempts at Language Transformation

Historians note 17th-century efforts

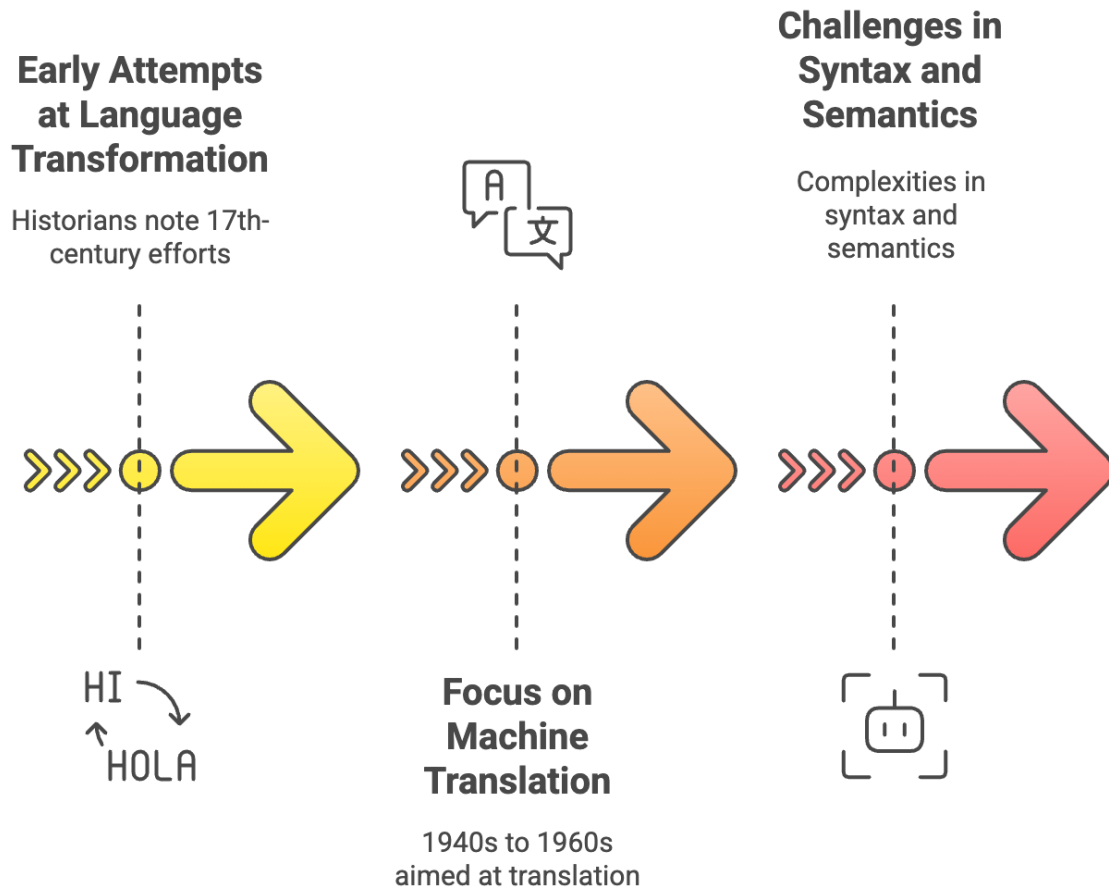


Focus on Machine Translation

1940s to 1960s
aimed at translation

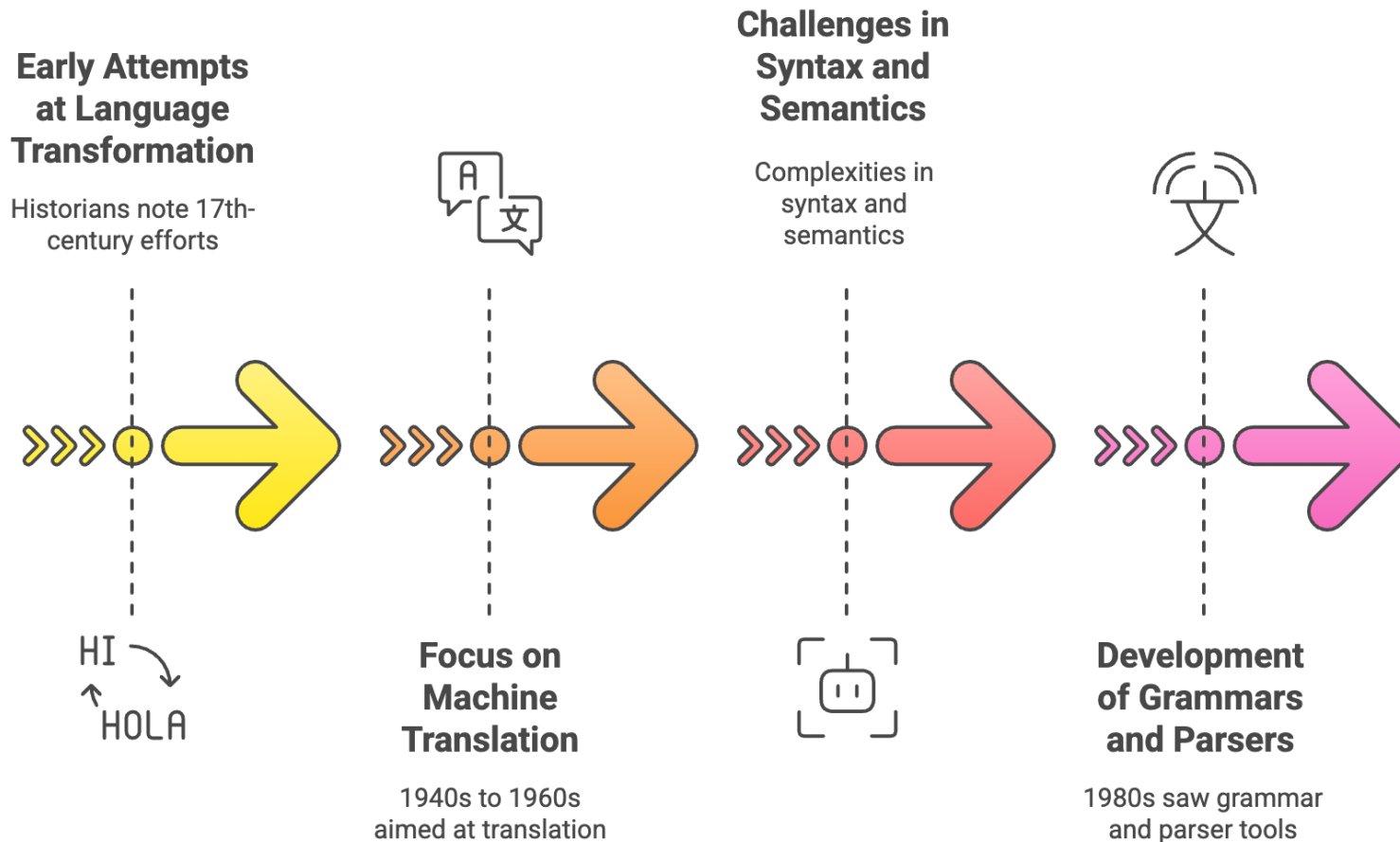
- Early approaches relied on word-to-word dictionary translations, where each word was looked up in a bilingual dictionary and replaced with its counterpart.
- **Challenges:**
 - This method was slow and inefficient.
 - It failed to capture sentence structure, syntax, and meaning (e.g., idioms and context-dependent words).

THE HISTORY OF NLP



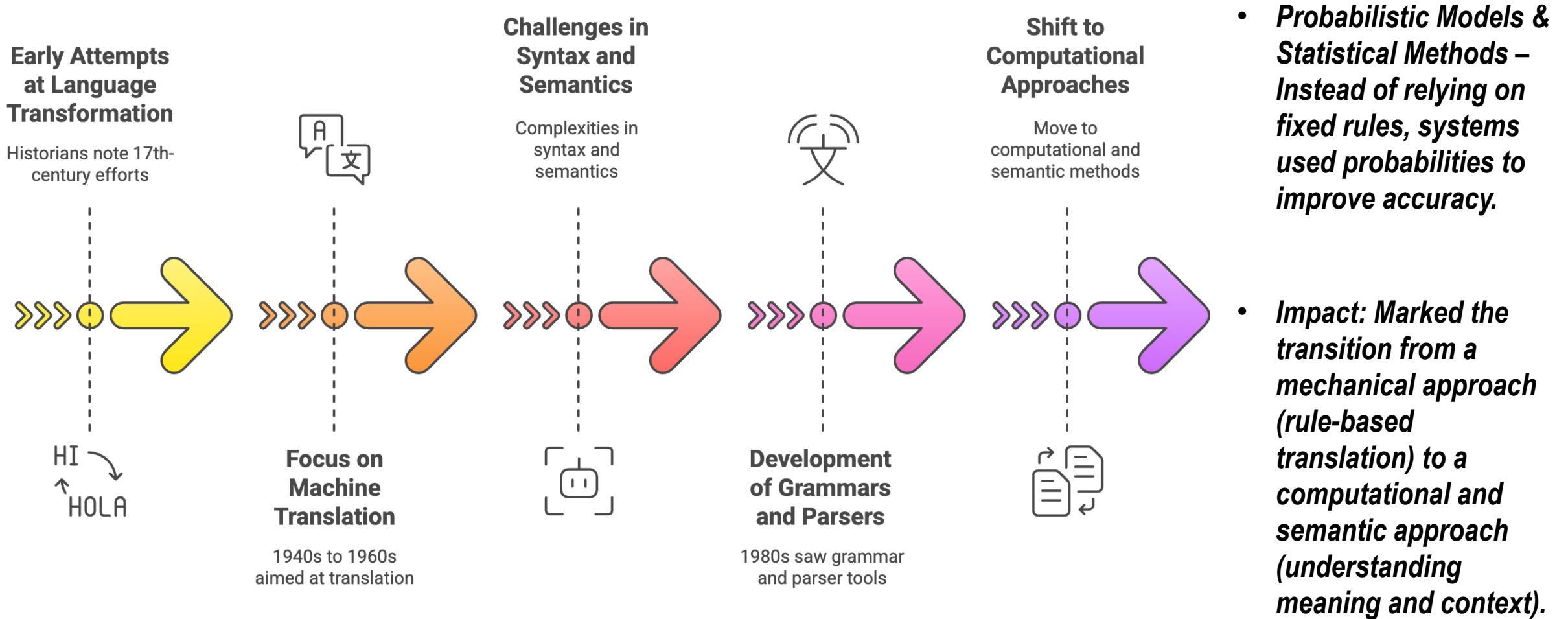
- ***Syntactic processing (understanding sentence structure).***
- ***Semantic processing (understanding meaning and context).***
- ***Direct word-for-word translation often led to incorrect or unnatural sentences.***

THE HISTORY OF NLP

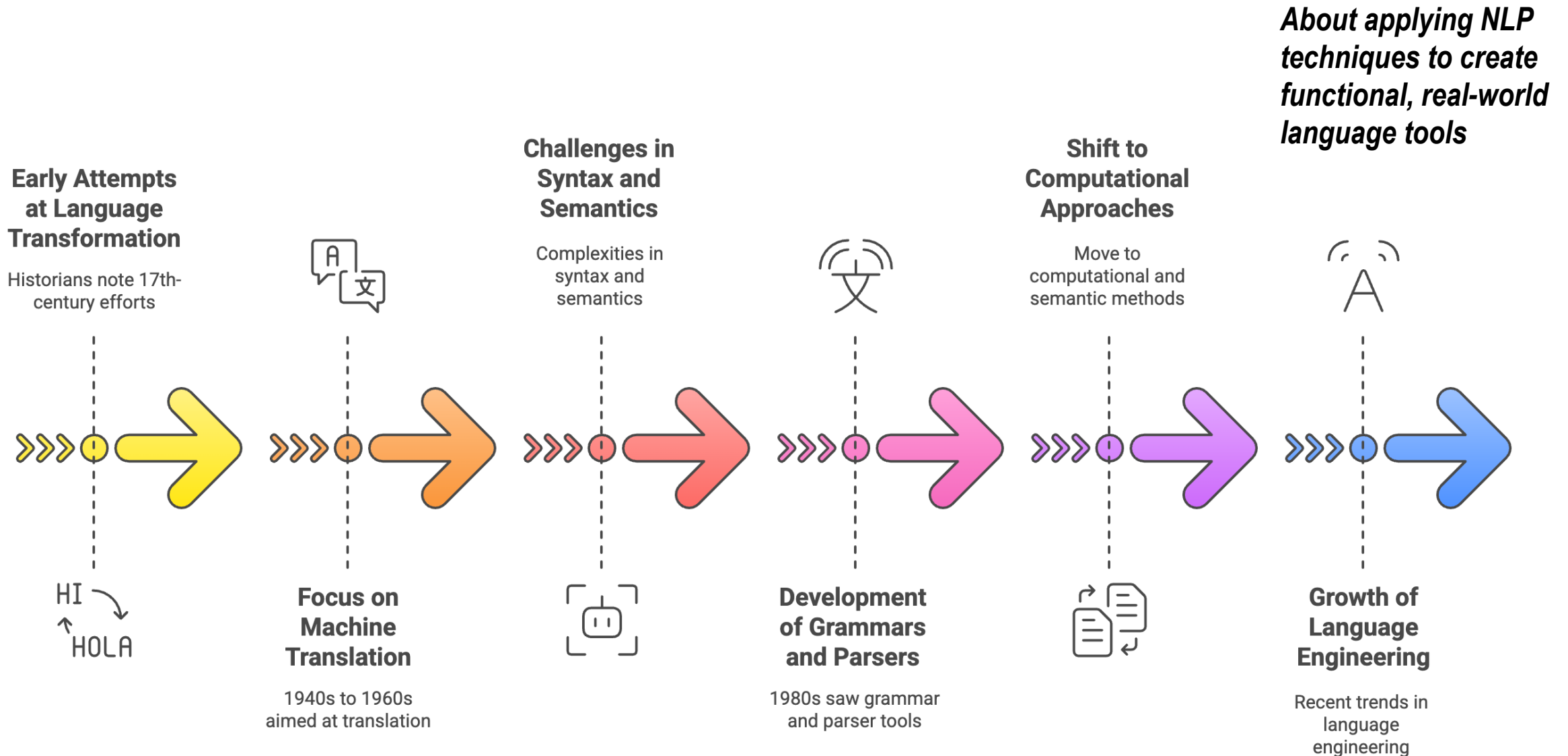


- **Parsers & Grammars** – Systems began analyzing not just words but also sentence structure and rules.
- **Word Sense Disambiguation (WSD)** – Understanding words with multiple meanings (e.g., "bank" as a financial institution vs. "bank" of a river).

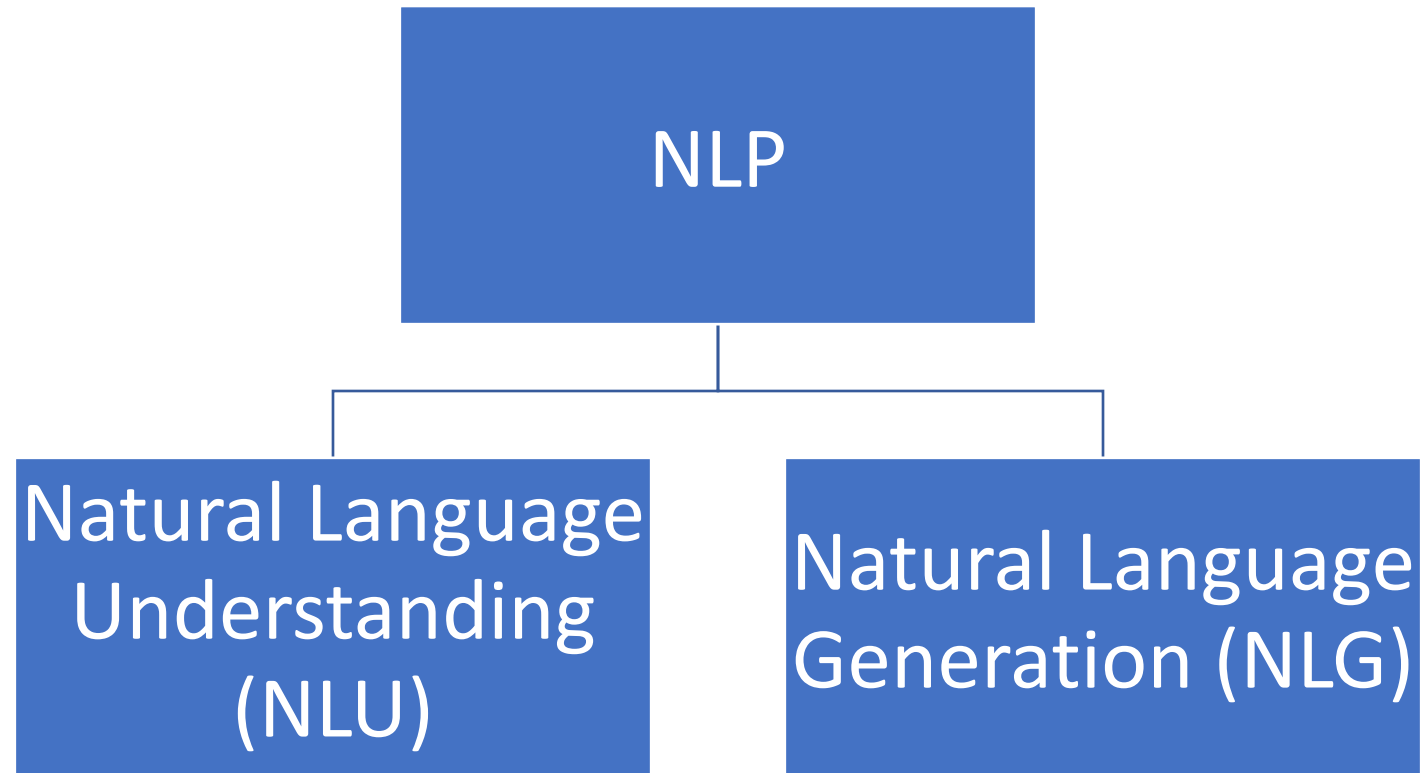
THE HISTORY OF NLP



THE HISTORY OF NLP



Broad Categories of NLP



NLU refers to a process by which an inanimate object with computing power is able to comprehend spoken language

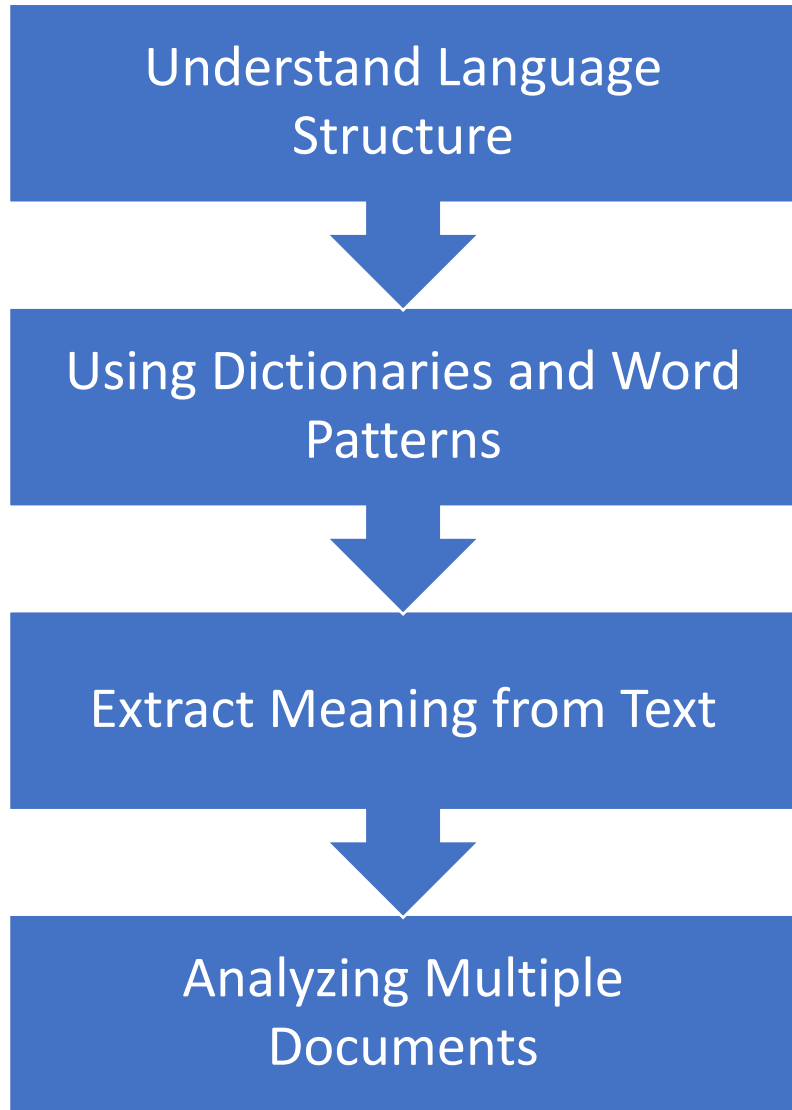
NLG refers to a process by which an inanimate object with computing power is able to manifest its thoughts in a language that humans are able to understand

Text Analytics vs NLP

- Text Analytics - The art of extracting useful insights and answering questions from any given text data
- NLP – Not just restricted to text data. Voice (speech) recognition and analysis also come under domain of NLP.



How does NLP work? General Framework



NLP systems analyze language by recognizing patterns, structures, and meanings in words and sentences.

- Example: "The cat is sleeping on the mat."
- NLP identifies "cat" as a noun, "sleeping" as a verb, and "on the mat" as location information.

Computers use dictionaries (word meanings) and patterns of how words appear together to understand context.

- Example: "Apple is a fruit." 🍏 → Here, Apple refers to food.
- "Apple released a new iPhone." 📱 → Here, Apple is a company.
- NLP can recognize the difference using context!

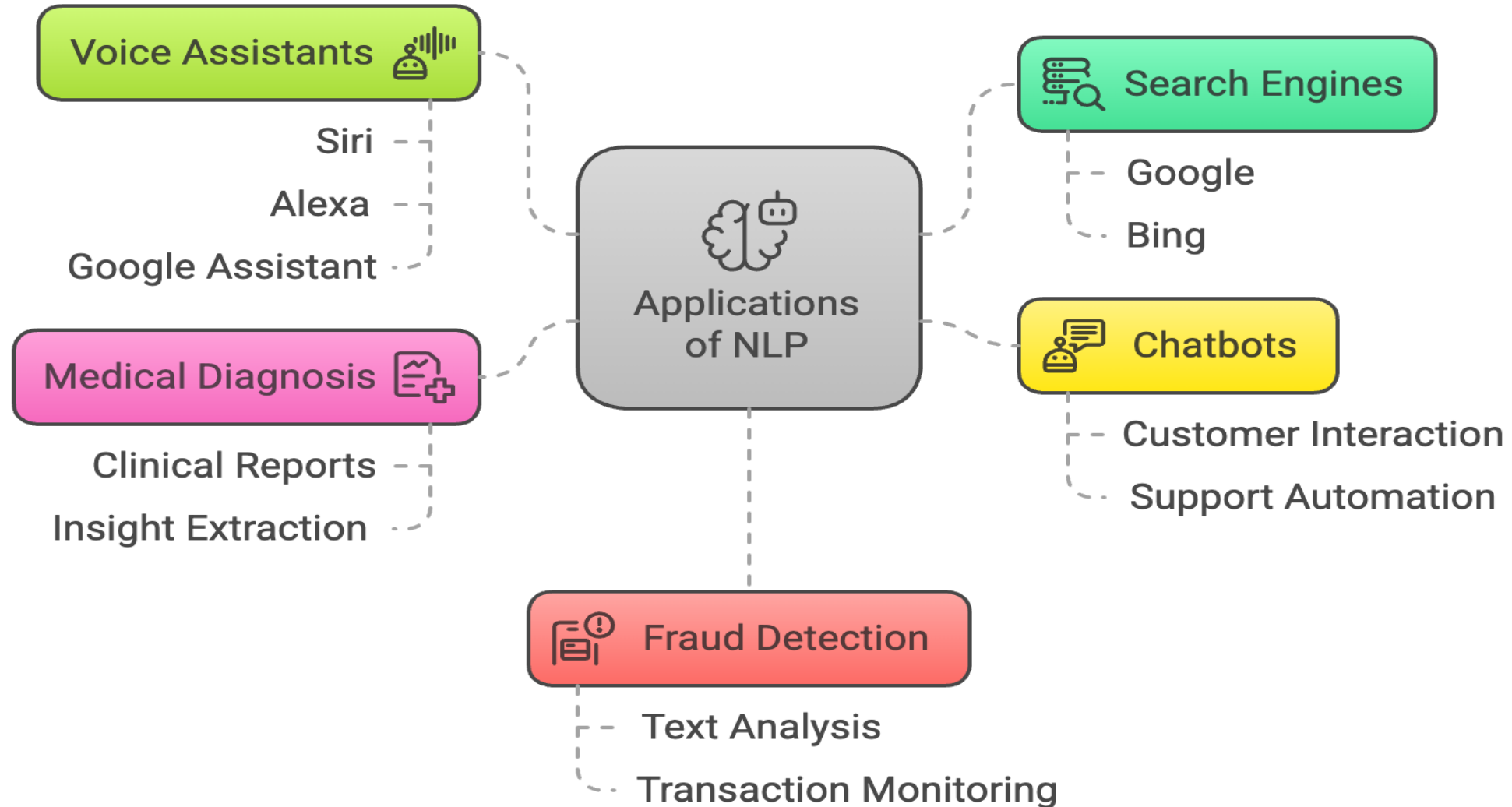
NLP can find important information like names, locations, events, and actions from a document. Example: If you give a computer a news article about "Elon Musk launching a SpaceX rocket from Florida", NLP can extract:

- Person: Elon Musk; Company: SpaceX; Action: Launching; Location: Florida
- NLP is not limited to one sentence—it can compare and analyze multiple documents!
- Example: If two news articles mention "the same accident but use different wording", NLP can still identify it's the same event.

Example

London is the capital and most populous city of England and the United Kingdom. Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia. It was founded by the Romans, who named it Londinium. London's ancient core, the City of London, largely retains its 1.12-square-mile (2.9 km²) medieval boundaries.

Applications of NLP



Understanding Linguistics

- NLP = Linguistics + ML + Computational Techniques
- Linguistics – The Study of Language
 - Linguistics is the scientific study of language and its structure, including how it is formed, used, and understood.
 - It explores both the rules that govern languages and how humans acquire and process them.

Branches of Linguistics/NLP Terminology

Replaying" → "re-" (prefix) + "play" (root) + "-ing" (suffix).

Morphology

Study of the structure of words

Syntax

Study of sentence structure

"The cat chased the mouse" vs. "The mouse chased the cat" (same words, different meanings).

Semantics

Study of meaning in language

Pragmatics

Study of language in context

"Can you open the window?" is a request, not just a question about ability.

Phonology

Study of how sounds function in a language

Phonetics

Study of speech sounds and their physical properties

In English, "ng" appears at the end of words (e.g., "sing") but never at the beginning.

difference between pronouncing "p" (with a puff of air) and "b" (without a puff of air).
pitch, loudness, frequency



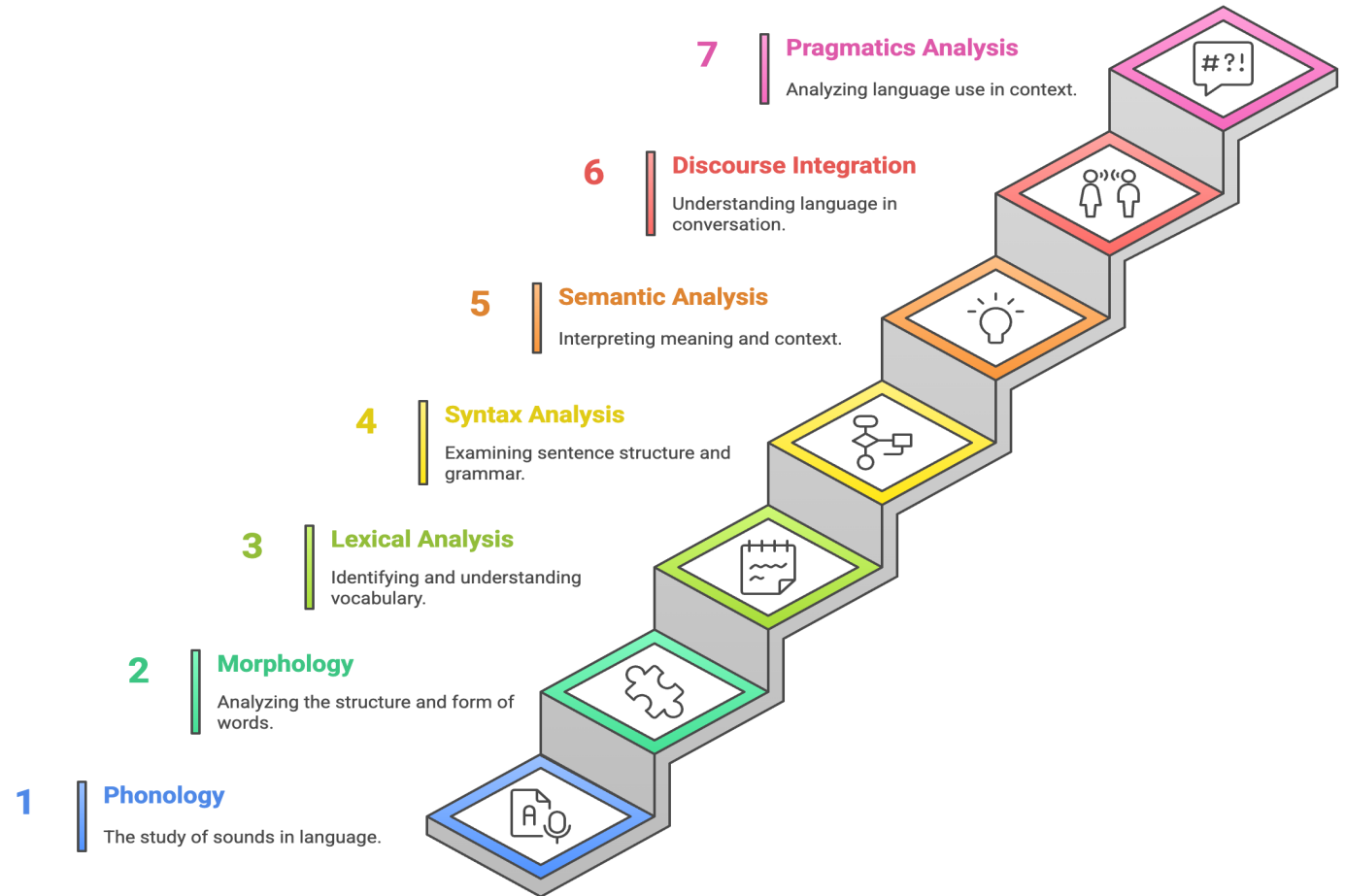
Role of Linguistics in NLP

Role in NLP :

- Linguistic analysis breaks down the text to provide meaning.
- Linguistics provide rules and structures in NLP that help AI systems understand and generate human language.
 - For example:
 - Syntax & Parsing help computers understand sentence structure.
 - Semantics improves meaning extraction in chatbots.
 - Phonetics & Speech Processing aid in voice recognition technologies.

Main Steps/Phases/Levels of NLP

1. Phonology
2. Morphology
3. Lexical Analysis
4. Syntax Analysis
5. Semantic Analysis
6. Discourse Integration
7. Pragmatics Analysis



Phonology in NLP

- **Phonology is the study of the physical sounds of a language and how those sounds are uttered in a particular language.**
- **This area is important for speech recognition and speech synthesis but is not important for interpreting written text.**
- **However, to understand, for instance, the soundtrack of a video, or the recording of a call center call, not only is the pronunciation of the words important (regional accents such as British English or Southern United States), but the intonation patterns.**
- **A person who is angry may use the same words as a person who is confused; however, differences in intonation will convey differences in emotion.**

Phonology in NLP

- **Phonemes** – The smallest unit of sound that can change meaning.
 - Example: In English, **/p/** and **/b/** are phonemes because they distinguish words like *pat* and *bat*.
- Why is Phonology important?
 - Helps in understanding **speech patterns** and **accents**.
 - Essential for **language learning** and **speech therapy**.
 - Useful in **speech recognition** and **text-to-speech** systems (e.g., Siri, Google Assistant).

Morphology

- Morphology refers to the structure of a word.
- Morphology breaks words into morphemes (smallest units of meaning).
- Types of Morphemes:
 - **Root Morpheme:** *The base word (e.g., "happy" in "unhappiness").*
 - **Affixes:** *Prefixes and suffixes that modify meaning (e.g., "un-" (prefix) and "-ness" (suffix) in "unhappiness").*

Step 1. Lexical Analysis/ Text Preprocessing

- In any analysis of incoming text, the first process is to identify which language the text is written in and then to break the text apart into separate sentences.
- Later, separate the sentences into words (Tokenization).
- **Step 1.1 Tokenization: Process of splitting a sentence into its constituents i.e. separate words or tokens.**
- Eg. “I am reading a book.” -----> “I”, “am”, “reading”, “a”, “book”, “.”

n-gram refers to a sequence of n items from a given text.

- Unigrams – One token at a time
- Bigrams- Extract two tokens at a time
- Trigrams – Extract three tokens at a time

These tokens are called Unigrams (one token at a time)

Eg: I would love to visit the United States

Unigram can give wrong results here as United States must be treated as single entity.

Lexical Analysis – Issues with Tokenization

1. Ambiguities that arise due to whitespaces, hyphens, contractions, etc.

- Many languages like Chinese and Japanese do not separate words with spaces.

Chinese Text:

我喜欢看电影 (I like watching movies)

Possible tokenizations:

- ["我" (I), "喜欢" (like), "看" (watch), "电影" (movie)] -> Right
- ["我" (I), "喜欢看" (like watching), "电影" (movie)] -> Wrong

- Multi-word expressions(MWE)

English Phrases:

"New York" may be split into ["New", "York"] instead of ["New York"].

Idioms:

"Kick the bucket" (meaning "to die") should not be split into ["Kick", "the", "bucket"].

Lexical Analysis – Issues with Tokenization

1. Ambiguities that arise due to whitespaces, hyphens, contractions, etc.

- Contractions.
 - **English Contractions:** The appropriate choice depends on the tokenization rules and the NLP application.
 - *"don't"* → Should it be ["do", "n't"] or ["don't"]?
 - *"I'm"* → Should it be ["I", "'m"] or ["I'm"]?
- Punctuation Issues
 - **Sentence:** *"Dr. Smith is a Ph.D. holder."*
Incorrect: ["Dr", ".", "Smith", "is", "a", "Ph", ".", "D", ".", "holder", "."] ❌
Correct: ["Dr.", "Smith", "is", "a", "Ph.D.", "holder", "."] ✅
- Numbers and Dates
 - **Date:** *"12/05/2024"*
["12", "/", "05", "/", "2024"] ❌
["12/05/2024"] ✅
 - **Number:** *"\$10.5 million"*
["\$", "10", ".", "5", "million"] ❌

- For example, if you are building a rock band search engine, you want to make sure you don't ignore the word "The". Because not only does the word "The" appear in a lot of band names, there's a famous 1980's rock band called *The The*!

How to customize stop words for specific applications?



Generic Stop Words

Use a standard list for broad applications



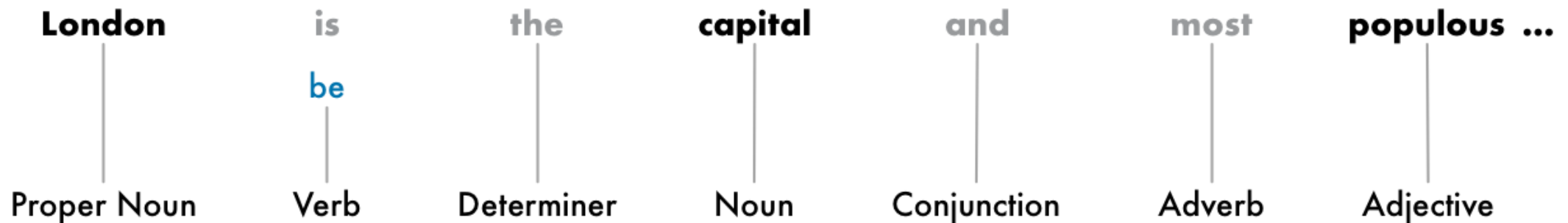
Context-Specific Stop Words

Tailor stop words to specific needs

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.2: StopWord Removal (Later -> Zipf's Law)

- Stop words - Common words that are just used to support construction of sentences
 - Do not impact meaning of sentence and can add noise
 - Remove the most commonly occurring words that don't add much information to the text. For example, "the," "a," "an," and so on.
 - There's no standard list of stop words that is appropriate for all applications



Stop words grayed out

Another example: I am happy vs I am not happy

Statistical Nature of Languages

- We now know : *how text can be processed linguistically (like lexical-> stemming/lemmatization -> syntactic -> semantic ->.....)*

Can we understand text statistically?

- **Sentence 1:** The cat is sitting on the mat.
- **Sentence 2:** The cat is sleeping on the mat.

Words repeated: the, is, on, mat, cat

Some words change: sleeping, sitting

Words occur with different frequencies.

NLP makes extensive use of this statistical property.

Word Frequency

- Number of times a particular word occurs in a text or corpus
- Example Corpus:
 - *D1: AI is useful.*
 - *D2: AI is powerful.*
 - *D3: AI helps humans.*
- *Example 2:* Suppose a document with 10,000 words
 - The word: "**the**" occurs 800 times.
 - The word: "**quantum**" occurs 15 times.



Word	Frequency
AI	3
is	2
useful	1
powerful	1
helps	1
humans	1

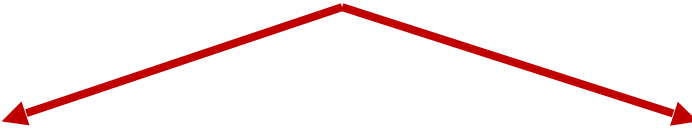
Does a high frequency automatically mean that a word is important?

NO!

Token vs Type

- **Token** - an individual occurrence of a word in the text
- **Type** - a unique word

cat dog cat bird dog cat



Tokens – 6

Tokens: {cat dog cat bird dog cat}

Gives corpus length

Type – 3

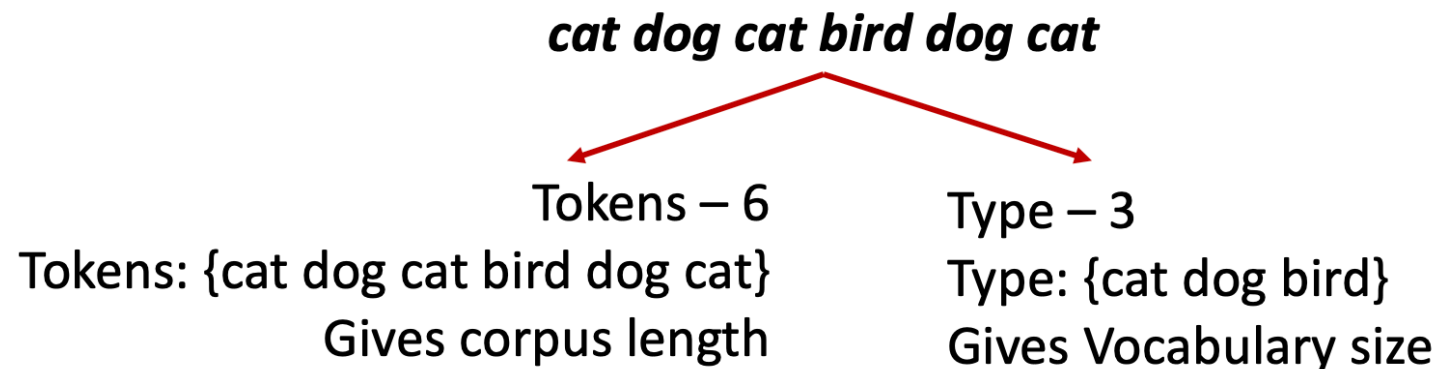
Type: {cat dog bird}

Gives Vocabulary size

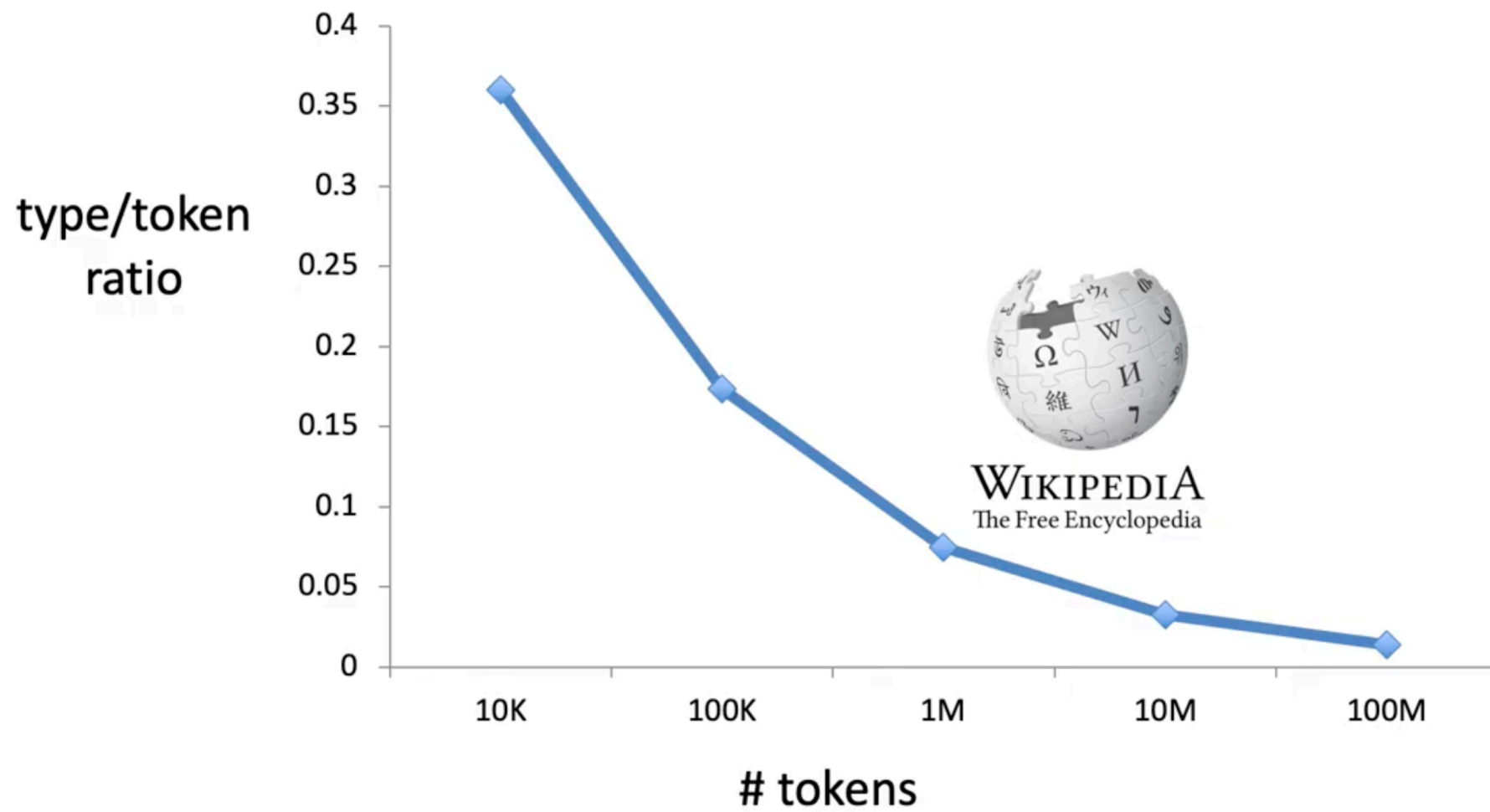
Why It Matters in NLP

- **Vocabulary Size (Types):** Tells the model how many distinct words it needs to learn.
- **Corpus Length (Tokens):** Measures the total size or volume of the text processed.
- **Type-Token Ratio (TTR):** Dividing types by tokens gives you the TTR, which helps measure text vocabulary richness and complexity.

- Here, $TTR = 3/6 = 0.5$



How does the type/token ratio change when adding more data?





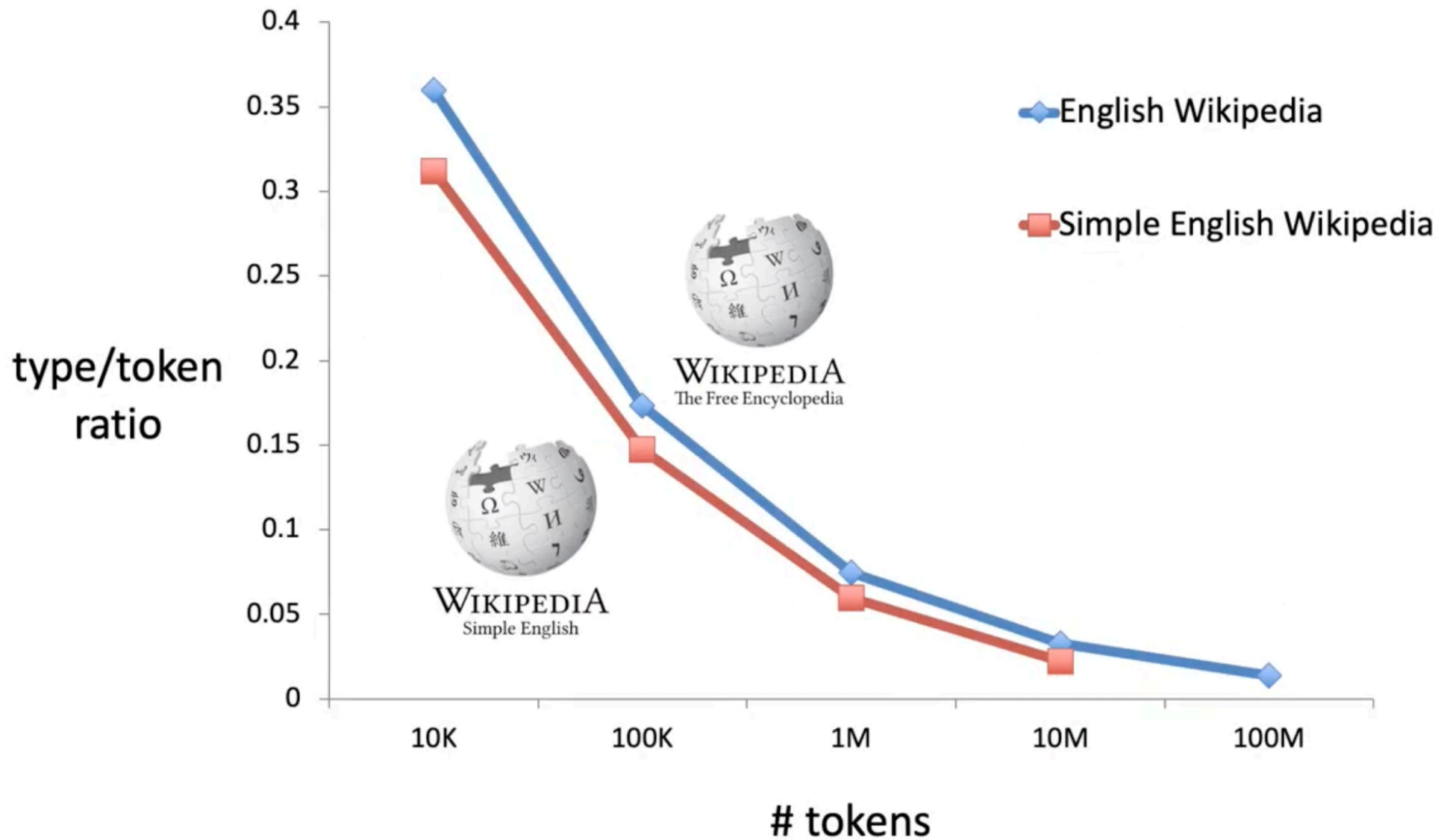
WIKIPEDIA
The Free Encyclopedia

vs.



WIKIPEDIA
Simple English

Which has a higher type/token ratio?



Higher TTR is a measure of complexity of the corpus and indicates greater lexical variety within the analyzed text.

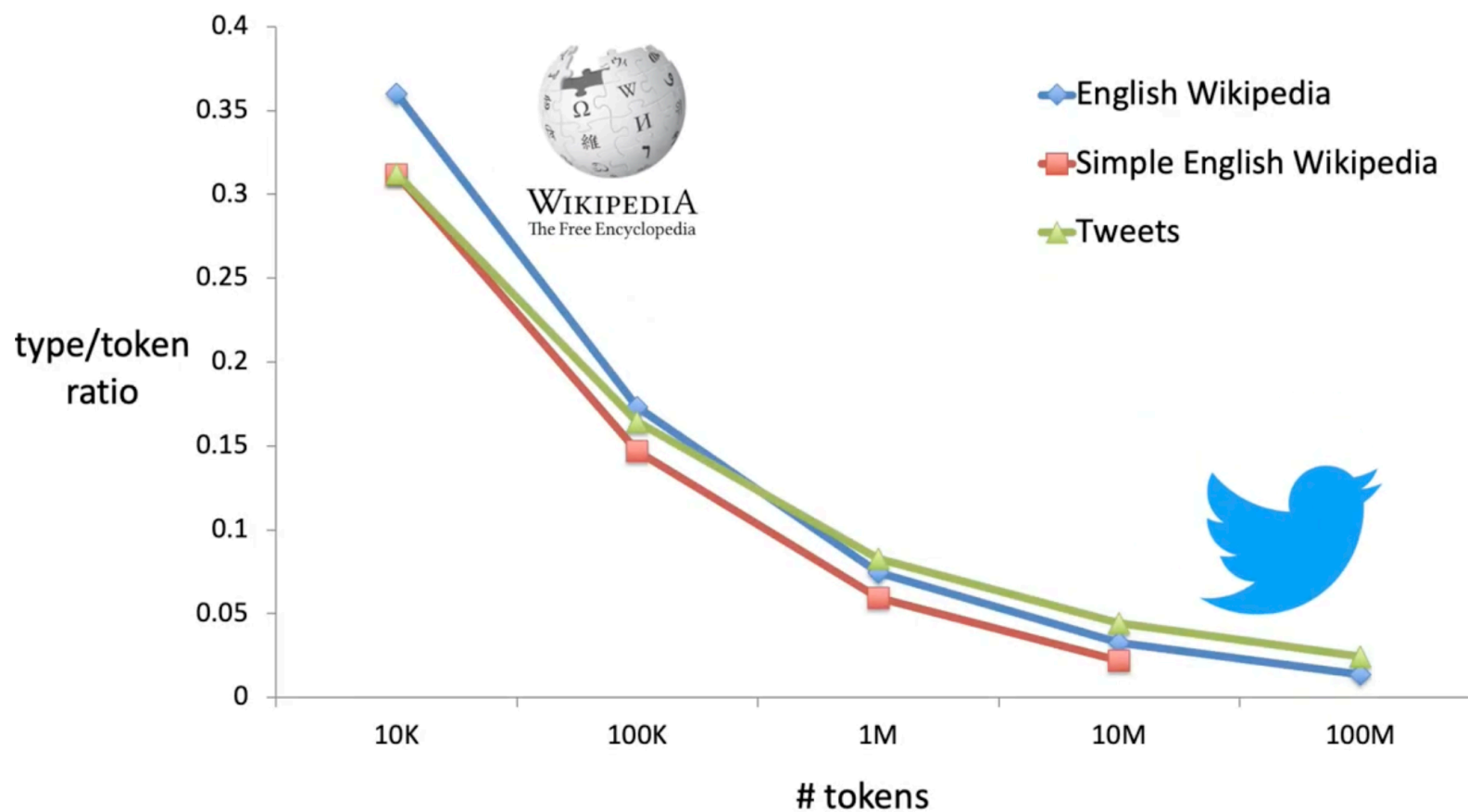


WIKIPEDIA
The Free Encyclopedia

VS.



For small dataset, twitter has smaller TTR... but its TTR increases as more data is added..



WHY?

224571 really	38 really2	12 reaaaaaally	7 reeeeeally	5 rrly	3 realiy
1189 rly	37 reaaaaaally	12 rreally	7 reeeeeeeeeaally	5 rrrreally	3 realllllllllllllllllly
1119 realy	35 reallyyyyy	11 reaally	7 relaly	4 reaaaaaly	3 reallllllyy
731 rlly	31 reely	11 reallllllyyy	6 r-e-a-l-l-y	4 reaaalllly	3 reallllllyyyy
590 realllly	30 reallllyyy	11 reeeallly	6 r-really	4 reaaallllyy	3 reallllllyyyyyyy
234 realllly	27 reaaly	11 reeeeeallly	6 reaaaaaaallly	4 reaalllly	3 reaalllyyyyy
216 reallyy	27 realllly	10 reaaaly	6 reallllllllllly	4 reaalllyyy	3 realluy
156 relly	26 reallllyyyy	10 reallyreallyreally	6 reallllyyyyy	4 reallllllllyyyy	3 really)
146 reallllly	25 reallllllllly	9 r)eally	6 realyl	4 reallllllyyy	3 reallyl
132 rily	22 reaaallly	9 really-really	6 reeeaaaally	4 reeeaaaally	3 reallyyyyyyyyyy
104 reallyyy	21 really-	9 reallys	6 reeeaaaallly	4 reeealy	3 reeeaaaallly
89 realllllly	19 reeaally	9 reeeeeeeally	6 reeeaaaalllyyy	4 reeeeeeeeeeeally	3 reeaalllly
89 reeeally	18 reallllyyy	8 realky	5 reaaaaaallly	4 rllly	3 reeaalllyyy
84 reaaally	16 reaaaallly	8 reallyyyyyyy	5 reaaaalllly	3 r3ally	3 reeaaly
82 reaally	15 reaallly	8 reallyyyyyyyy	5 reaalllly	3 rjeally	3 reeaallly
72 reeeeeally	15 reallllllllly	8 reeeeaally	5 reallllllllllllly	3 reaaaaaaaally	3 reealy
65 reaaaally	15 reallllyy	7 r3ally	5 realllllllllllly	3 reaaaaaaly	3 reeeaaaallllyyy
57 reallyyyy	15 reallyreally	7 raelly	5 reeallyyy	3 reaaaallllyy	3 reeeaaallly
53 rilly	15 realyy	7 reaaaaaaally	5 reeeaaaallly	3 reaaaalllyy	3 reeeeaallaally
50 realllllllly	14 reallllyyyy	7 reallllllllllllly	5 reeeeaally	3 reaalllyy	3 reeeeaalllly
48 reeeeeeeally	14 reeeeeeeally	7 reallllllyyy	5 reeeeeeeeeally	3 reaallllyy	3 reeeeealllly
41 reeally	13 reeeaaaally	7 reeeeeaaally			

- Because twitter shows enormous amounts of variations in word forms

How many words are there?

- Size of vocabulary continues to grow as you collect more data
- You will never find all the words
- How words are distributed? – Some many times, some only once



```
1 rreeeeeeaaaaallllllllyyyyyy
1 rreeeeeeallly
1 rreeeeeely
1 rrreallyyy
1 rreeeaaaallllyy
1 rreeeallllyy
1 rreeeaaalllllyy
1 rreeeallly
1 rrrllyy
1 rrrreeally
1 rrrreeeeeeaaaaallllllllyyyyyy
1 rrrreeeeeeaaallly
1 rrrreeeeeeaaalllyy
1 rrrrreally
1 rrrrrealy
1 rrrrrreeeeeeaaaaallllllyyyyyy
1 rrrrrrrreally
```

How words are distributed in natural language?

Zipf's Law in Linguistics

- Zipf's Law states that in a large corpus of natural language text, the ***frequency of a word is inversely proportional to its rank in the frequency table.***

- Mathematically, $f \propto \frac{1}{r}$

Or , $f = \frac{C}{r}$

- If we count up how often each word (type) of a language occurs in a large corpus, and then list the words in order of their frequency of occurrence, we can explore the relationship between the frequency of a word f and rank its position in the list, known as its *rank* r .

Suppose:

$$C = 10,000$$

Then:

$$f(r) = \frac{10,000}{r}$$

For rank 1:

$$f(1) = 10,000$$

For rank 2:

$$f(2) = 5,000$$

For rank 5:

$$f(5) = 2,000$$

For rank 10:



$$f(10) = 1,000$$

If the rank increases by a factor of 10, the frequency approximately decreases by a factor of 10.

Suppose the word at rank 20 occurs 500 times. What is the approximate frequency of the word at rank 100?

Using:

$$f(r) = \frac{C}{r}$$

At rank 20:

$$500 = \frac{C}{20}$$

Therefore:

$$C = 10,000$$

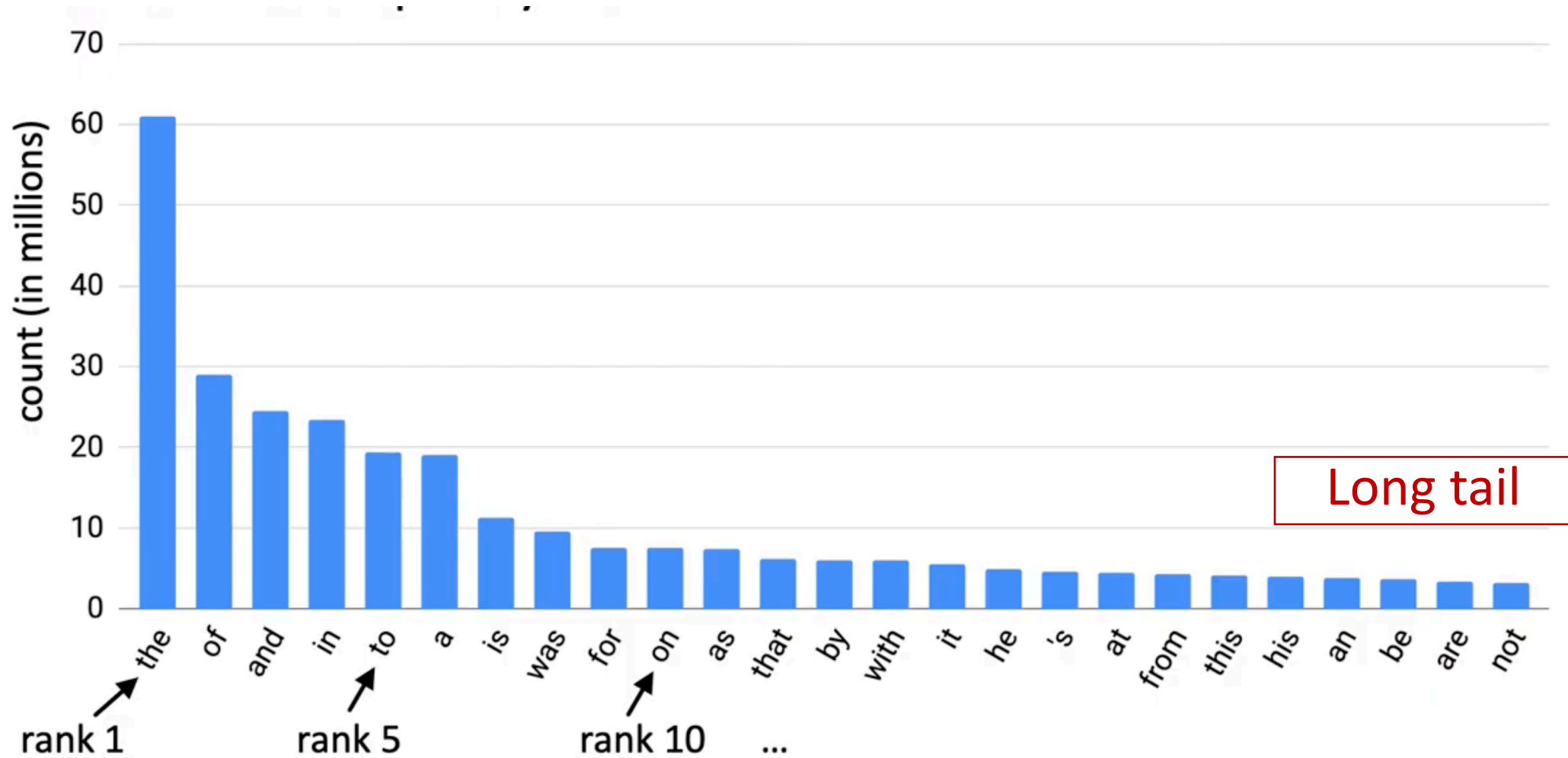
At rank 100:

$$f(100) = \frac{10,000}{100}$$

100

Zipf's Law is an **empirical approximation**.

Zipf's Law



Long Tail Challenges in NLP

- Majority of word types are very rare so difficult to get enough data for systems to learn how to handle them
- So many word types !
- But, words have **internal structure (in addition to spelling variations)**, words have morphological structure like suffixes, prefixes, etc.

play
plays
played
playing
player
players
replay
replays
gameplay
horseplay

Zipf's Law

- A few words occur extremely frequently.
- Many of these are function words.
- Some very frequent words may contribute little topic-specific information.
- This motivates techniques such as:
 - stop-word removal
 - frequency thresholding
 - TF-IDF weighting
- But remember: Zipf's Law does not say "remove stop words".
- It only describes **the frequency** distribution.
- But, frequency alone is a problem!

Summary

- to do NLP on some text, we need to preprocess it:
 - tokenize documents into sentences
 - tokenize sentences into tokens
- rule-based tokenizers exist for many languages
- for writing systems without whitespace, tokenization becomes complex (often treated as an NLP problem)

Summary

- useful terms: **type**, **token**, **type/token ratio**
- when adding data, number of types keeps increasing
- most types are extremely rare (Zipf's law)

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.3: Text Normalization

Process wherein different variations of text get converted to standard form

- **Spelling Correction**
- **Stemming**
- **Lemmatization**

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.3: Text Normalization – Spelling Correction

- Autocorrect – Python library
 - Natureal -----> Natural

```
sentence = word_tokenize("Ntural Luanguage Processin deals with the art of  
extracting insightes from Natural Languaes")
```

```
Natural Language Proceession deals with the art of extracting insights from Natural Languages
```



Need spelling
correctors which are
aware of context

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.3: Text Normalization - Stemming

Stemming - A text preprocessing technique that reduces words to their root or base form (stem) by removing affixes, simplifying text for tasks like information retrieval and text analysis

- The words "running," "runner," and "runs" would all be reduced to the stem "run"
- Running and runner ---→ runn (ing and er)
- Playing -> play; happily --→ happili
- **Problem with Stemming: Stemmed words do not carry any meaning**
 - e.g. : independence -> independ (which is not present in English Dictionary)
 - Lemmatization deals with such cases by using a vocabulary and analyzing the words' morphologies.
 - It returns the base forms of words which can be actually found in dictionaries

Step 1. Lexical Analysis/ Text Preprocessing

Step 1.4: Text Normalization - Lemmatization

Lemmatization - A text preprocessing technique that reduces words to their base or dictionary form (lemma) by considering the context and part of speech (POS); figuring out the most basic form or lemma of each word in the sentence.

- ‘running’ -> ‘run’
 - ‘better’ -> ‘good’
 - ‘went’ -> ‘go’
 - ‘rocks’ -> ‘rock’
 - Corpora -> corpus
- Lemmatization takes into account the grammatical category of a word (noun, verb, adjective, etc.) and provides the base form accordingly.
 - For example, the lemma of “running” as a verb is “run,” while as a noun, it remains “running.

Stemming vs Lemmatization

- **Stemming is an informal process of converting words to their base forms using heuristic rules. For example, “university,” “universities,” and “university’s” might all be mapped to the base univers. (One limitation in this approach is that “universe” may also be mapped to univers, even though universe and university don’t have a close semantic relationship.)**
- **Lemmatization is a more formal way to find roots by analyzing a word’s morphology using vocabulary from a dictionary.**
- **Stemming and lemmatization are provided by libraries like spaCy and NLTK.**
- **Lemmatization considers the context and morphological analysis to return a valid word, whereas stemming applies simpler rules to chop off prefixes or suffixes, often resulting in non-dictionary words.**

Lexical Analysis

London is the capital and most populous city of England and the United Kingdom. Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia. It was founded by the Romans, who named it Londinium.

Sentence Segmentation



1. "London is the capital and most populous city of England and the United Kingdom."

2. “Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia.”

3. "It was founded by the Romans, who named it Londinium."

Tokenization



*“London”, “is”, “the”, “capital”, “and”, “most”, “populous”,
“city”, “of”, “England”, “and”, “the”, “United”, “Kingdom”,
“ ”
.”*

- <https://www.deeplearning.ai/resources/natural-language-processing/>